



JD for Agentic AI Instructor - Interview Kickstart

About Us

The name Interview Kickstart might have given you a clue. But here's the 30-second elevator pitch - Interviews can be challenging. And when it comes to the top tech companies like Google, Apple, Facebook, Netflix, etc., they can be downright brutal. Most candidates don't make it simply because they don't prepare well enough. Interview Kickstart (IK) helps candidates nail the most challenging tech interviews.

To keep up with the upcoming trends, we are launching our new Agentic AI course.

Requirements:

The Instructor must perform a hands-on demo on one of the following topics. For example, if someone is choosing to build a multi-agent project using MCP, the MCP framework can be discussed using slides, but please include a demo project where an MCP is built using a specific tool.

1. Fundamentals of Agentic AI: Introduction to LLMs, RAG, and Agents (with Coding)

Introduction to Large Language Models (LLMs): architecture, context, and prompting

Understanding RAG (Retrieval-Augmented Generation): embeddings, vector search, and document retrieval

Connecting RAG with Agents for context-aware reasoning and action

Implementing basic RAG and agent pipelines in Python using LangChain or LlamaIndex

2. Building Applications with LLMs and Agents

Designing end-to-end applications integrating LLMs, APIs, and external tools

Building multi-step reasoning workflows using agent memory and planning

Integrating guardrails, safety checks, and content moderation

Deploying LLM-powered applications with FastAPI, LangChain, or Streamlit



3. Build Your First Agent with Python: LangChain

Introduction to LangChain architecture: LLMs, Tools, Memory, and Chains

Building a modular LangChain Agent capable of reasoning and decision-making

Implementing tool invocation, function calling, and conversational memory

Creating a functional, deployable LangChain-based agent in Python

4. Build a Multi-Agent System with CrewAI / AutoGen and LangGraph

Understanding collaborative agent frameworks: CrewAI, AutoGen, and LangGraph

Designing agent roles for planning, reasoning, and task delegation

Implementing graph-based agent workflows and inter-agent communication

Deploying a multi-agent system that autonomously completes complex tasks

5. Build Multi-Agents with Model Context Protocol (MCP) and Agents-to-Agents Protocol

Introduction to MCP and A2A Protocol for structured multi-agent communication

Defining shared context, message passing, and goal synchronization between agents

Building cooperative workflows where agents exchange reasoning and results

Implementing agent-to-agent communication pipelines using Python and LangGraph

6. Conversational Audio Bots or Agents with Multimodal Data

Integrating speech, text, and vision into unified multimodal agent workflows

Building conversational audio agents using ASR (Whisper) and TTS systems

Combining audio understanding with LLM reasoning for task automation

Deploying real-time multimodal conversational agents using GPT-4o or OpenAI Realtime APIs

7. Specialized Agents

Designing domain-specific agents for specialized use cases (e.g., data, healthcare, coding, support)

Integrating APIs and tools tailored to domain-specific tasks and reasoning

Implementing modular architectures for focused, high-accuracy agent behavior



Deploying specialized agents that perform targeted, expert-level tasks autonomously

8. Evaluating and Optimizing Agents with Deployment using LangSmith or other tools

Monitoring agent performance using LangSmith dashboards and traces

Optimizing inference latency, model selection, and token costs

Comparing fine-tuning vs. prompt optimization for improved outcomes

Deploying and continuously evaluating agents for accuracy, cost, and reliability

9. Python for GenAI

Refresher on Python foundations essential for GenAI workflows

Using key libraries: OpenAI, LangChain, FastAPI, Pydantic, and Vector Databases

Working with APIs, JSON data, and async workflows for agent development

Building and testing Python pipelines for LLMs, RAG, and Agents

Preferred Qualifications:

- Prior experience building and deploying LLM or agent-based applications in real-world settings
- Strong proficiency with agent frameworks like LangGraph, CrewAI, or LangChain (code-based for SWEs and low-code or no-code for Tech Domains)
- Strong understanding of system design principles, especially in AI/ML-based architectures
- Demonstrated ability to explain complex technical topics to diverse audiences
- Experience teaching, mentoring, or creating content for working professionals in tech
- Excellent communication and collaboration skills, with a learner-first mindset
- Bonus: Contributions to open-source AI projects, publications, or prior experience with AI upskilling programs

Responsibilities:



- **Instruction Delivery:** Conduct lectures, workshops, and interactive sessions to teach machine learning principles, algorithms, and methodologies. Instructors may use various teaching methods, including lectures, demonstrations, hands-on exercises, and group discussions.
- **Industry Engagement:** Staying current with the latest trends and advancements in machine learning and related fields, engaging with industry professionals, and collaborating on projects or internships to provide students with real-world experiences.
- **Research and Development:** Conducting research in machine learning and contributing to developing new techniques, models, or applications.
- Constantly improve the session flow and delivery by working with other instructors, subject matter experts, and the IK team.
- Help the IK team in onboarding and training other instructors and coaches
- Have regular discussions with IK's curriculum team in evolving the curriculum.
- Should be willing to work on weekends/evenings and be available as per the Pacific time zone

Demo Guidelines for Agentic AI - Part-time Instructor

Objective:

To assess the candidate's technical expertise in Agentic AI and ability to communicate and teach complex topics effectively.

Note: *Please read the assignment details thoroughly before attempting the demo. Adherence to guidelines is one of the critical evaluation criteria.*

Assignment Details:

Prepare a technical demo presentation on one of the topics provided to you by the recruiter. The presentation should be approximately 30 minutes long and showcase your technical understanding and teaching skills. *You can extend your demo duration for up to 45-60 minutes if you choose a difficult/lengthy topic with more than 15-20 slides in the material provided.*



Evaluation Criteria:

- **Technical Expertise:**
 - Depth and accuracy of technical content.
 - Ability to explain complex concepts clearly and effectively.
 - Use of relevant and up-to-date examples or case studies.
- **Teaching Skills:**
 - Clarity and organisation of the presentation.
 - Ability to engage and interact with the audience.
 - Use of effective teaching aids and materials.
- **Presentation Skills:**
 - Confidence and professionalism in delivery.
 - Clarity and organisation of the presentation.
 - Ability to engage and interact with the audience.
 - Use of effective teaching aids and materials.

Student Persona

- Your students are working professionals with an average experience of ≥ 5 years who are looking to upskill themselves in the Generative AI field and start new roles in their dream companies(tier-1) in the machine learning domain.
- Their existing work experience can be in the Machine Learning field or any non-ML technical domain, such as Frontend/Backend engineering, DevOps, etc.
- You can assume their prior knowledge basis the topic you are creating the lesson plan/demo for. Do remember to specify your assumptions at the beginning of the demo

Demo Options

You can choose to demo your skills through **any one of the options below**:

- **Live demo** with the internal evaluation team
- **Demo video submission** followed by a short call with the hiring team upon selection



Note: Candidates can create their presentation or use one of the presentations provided by IK to ensure they cover the necessary points. They can also utilize any publicly available presentations with due credits.

Live Demo Instructions:

To help us assess both your technical depth and ability to teach effectively, please prepare a 15–20 minute live demo based on one or more topics in your area of expertise. Based on the nature of your topic, follow the corresponding guidelines below:



For Theoretical / Conceptual Topics

(E.g., Agentic AI Foundations, ReAct, Multi-Agent Systems, Prompt Engineering, AI System Design Principles)

- Prepare a slide deck (8–12 slides) that explains the key concepts clearly and concisely.
 - Use real-world analogies, industry applications, or case studies to make the explanation relatable and engaging.
 - Include 1–2 interactive elements, such as:
 - A quiz question or thought experiment
 - A "What would you do?" scenario or decision tree
 - Bonus: Briefly walk through how the concept integrates into real-world applications or downstream workflows (e.g., where prompt engineering fits in when building agents).
 - You may optionally show a simple code snippet, diagram, or tool interface to demonstrate practical relevance.
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For Project-Based / Application Topics



(E.g., Building AI Agents, Multi-Agent Systems, AI Assistants, LangGraph Workflows, Agent Evaluation & Optimization)

- Present a working prototype, code walkthrough, or interactive demo of an AI agent or system you've built or adapted.
- You may also use mockups or low-code tools to demonstrate the core idea if building a complete system is not feasible.
- The live demo should cover:
 - Problem Statement & Context: What real-world task is this agent/system solving?
 - Architecture Overview: Key components like planner, executor, tools, memory, APIs, etc.
 - Working Demo: Show the agent in action (or walk through code with sample inputs/outputs).
 - Key Design Decisions: Why this architecture/framework/tool? How did you enable reasoning or collaboration?
 - Challenges & Learnings: Any hurdles faced and how they were resolved.
- Include a visual diagram or LangGraph/CrewAI flow, even if briefly, to make the system easy to follow.

Examples of Demos (per Topic Area)

Topic Area	Sample Demo Ideas
ReAct Framework	Slide + code snippet using LangChain or AutoGen
LangGraph	Visual + working graph with step-by-step reasoning
CrewAI Multi-Agent System	Live run of agents collaborating on a workflow (e.g., task breakdown)
Prompt Engineering & Function Calling	Notebook showing structured prompts + function calls



AI System Design	Slide-based walkthrough with architecture diagram
AI Agent Evaluation	Metrics dashboard or logging system showcasing agent feedback loop
PM/TPM/EM Use Cases	Low-code mockup using Flowise, Zapier AI, or Relevance AI

Demo Video Submission Instructions:

- Record your presentation as a video and submit it in a commonly used format (e.g., MP4, AVI).
- Ensure the video quality is clear and the audio is audible.
- Submit the video and any supplementary materials (slides, code snippets, whiteboard) used in the presentation.
- Videos and other material will be used solely for internal evaluation purposes and will not be shared or used commercially.

Presentation Structure Guidelines

1. Welcome and Introduction (5 minutes):

- Greet the participants and briefly introduce yourself.
- Outline the agenda for the session and set expectations for participation.
- Provide a brief overview of the chosen topic and its relevance.

2. Main Presentation (15 minutes - extend it up to 25 mins if you submit a demo video):

- Deliver the technical content in a clear, logical sequence.
- Use slides or diagrams to illustrate key points and maintain visual engagement. You can also use whiteboarding to explain concepts more intuitively.
- Include practical examples or case studies to demonstrate the application of concepts.
- Highlight recent advancements or research findings relevant to the topic.

2. Interactive Elements (10 minutes - applicable for live demo):



- Conduct a live coding demonstration or interactive exercise.
- Encourage participants to ask questions or share their insights.
- Address questions in real time and provide additional explanations as needed.
- Use engagement tools like polls or quizzes to keep the audience involved.

2. Conclusion and Wrap-Up (2-3 minutes):

- Summarize the key points covered in the session.
- Provide additional resources or references for further learning.
- Thank the participants for their time and participation.
- Invite any final questions and provide closing remarks.

Example Presentation Structure:

Topic: Transformer Architecture and Attention Mechanism

1. Introduction (5 minutes):

- Introduce yourself and provide a brief background in AI.
- Explain the importance of the Transformer Architecture in modern AI.

2. Technical Content (20 minutes):

- A detailed explanation of the Transformer model, including its encoder and decoder components.
- Deep dive into the attention mechanism, explaining how it works and why it's crucial.
- Use diagrams to illustrate the flow of information through the Transformer model.
- Discuss practical applications of Transformer models, such as in NLP tasks.

2. Interactive Element (5 minutes):

- Live coding session demonstrating the implementation of a simple Transformer model using a framework like PyTorch or TensorFlow.
- Q&A session addressing potential questions about the Transformer model.

By preparing and delivering this demo assignment, candidates will have the opportunity to demonstrate their mastery of the subject matter and ability to communicate and teach complex Generative AI concepts effectively.



Sample Class Snippets at IK

- [Graph Algorithms for Technical Interviews | Omkar Deshpande | IK UpLevel MicroClass](#)
- [Problem Identification | UI & DOM for #FAANG+ Interviews](#)